

Advanced Deep Learning-Based Approach for Tooth Detection, and Dental Cavity and Restoration Segmentation in X-Ray Images

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Abstract—In recent years, much research has been focused on dental problem classification, detection, and segmentation using Deep Convolutional Neural Networks (CNN or DCNN) models. Early detection of oral issues is crucial to maintain oral health and prevent the need for invasive treatments. Dental radiographs assist in this by providing detailed images. Integrating intelligent methods with conventional imaging techniques can result in more accurate diagnoses and treatment plans for patients, dental professionals, and service providers. Our study presents a novel unified method that utilizes CNNs to segment dental restorations and cavities in X-ray images. Various datasets, including publicly available one annotated in this study, were used to evaluate the method's performance. The first step in our method is to detect teeth, identifying each tooth's number and location in accordance with the Universal Teeth Numbering (UTN) system. This study has also presented a new post-processing technique to address teeth ordering predicting issues in X-ray images. The detection model achieved high precision and recall metrics of over 90% after post-processing, outperforming existing methods. The second step aims to crop images of each tooth from the initial network and precisely segment the cavity and restoration. The segmentation model achieved high precision and recall of over 60%. We proportionately projected the predicted outcomes onto the original image for the final evaluation. The overall performance of our proposed method exceeded that of state-of-the-art models. This method detects abnormalities that may go unnoticed by humans, reducing workload and minimizing bias in treatment assessment.

I. INTRODUCTION

Dental caries, a chronic infectious disease, is experienced by more than 90 percent of all adults in the United States. It can be categorized based on the extent of the lesion as normal, early-stage, moderate, or advanced [1]–[3]. One of the most common exercises in general dentistry practices is the diagnosis of dental caries [4]. Since dentists vary in their ability to identify and estimate the depth of caries lesions, which can be linked to factors such as radiograph quality, viewing conditions, the dentist's experience and skill, and the duration of each examination [2], [5], [6], creating a program for dentists to detect caries lesions and reduce diagnostic variations is critical, analyzing radiography images with Neural Networks (NN) can help dentists predict the need for future surgical treatments such as guided tissue regeneration, splitting, and sinus elevation [7]. The utilization of NNs in medicine can significantly enhance the quality of diagnosis. Ilesan et al. has

reported that implementing NNs has decreased the processing time of cone-beam computed tomography (CBCT) images [8]. On the other hand, implementing NNs-based methods can also save resources and increase efficiency by reducing the manual workflow between patients, radiologists, and dentists.

In recent years, there has been a significant increase and progress in Artificial Intelligence (AI) applied to dentistry. The works shown in Table I demonstrate the current advancements in this area. The incorporation of DL in dentistry research is a prominent area of study on the evolution of diagnosis, treatment planning, and patient care within the dental field.

Ezhov et al. [6] conducted an experiment in which they gathered 20 dentists and divided them into two groups. One group received assistance from an AI, while the other did not. The results indicate that the first group performed better overall. According to the report, the average time taken by dentists who received assistance in analyzing radiography images was significantly lower than that of their unaided counterparts.

Cantu et al. [11] employed NNs to segment caries lesions and compared their approach against a group of 7 dentists. The average f1-score achieved by dentists in the segmentation of dental caries lesions is 41%, but the f1-score attained by their model is 73%. These results suggest that using NNs may hold significant potential in mitigating inconsistencies and inaccuracies within the medical domain. Unsal et al. [17] developed a method to automatically segment possibly malignant oral diseases and oral cancers using multiple deep CNNs. In another research study, Andrade et al. [18] have demonstrated the feasibility of employing a U-Net model for the visual segmentation of dental biofilm. This approach enables patients to utilize an RGB camera to detect fundamental oral hygiene characteristics.

Figure 1 depicts the three primary types of 2D radiography used for caries detection. Bitewings are the most often used radiographic images for dental diagnosis [2], [12], [19], [20]. Bitewings are primarily used to find cavities and caries, especially ones under restorations. Another type of image is the periapical radiograph, which provides a comprehensive view of the tooth's crown, root, periapical regions, and adjacent alveolar bone. The third type is the panoramic radiograph, offering a broader view that encompasses not only the teeth

TABLE I: DL Approaches in Dentistry Research: Numbering (N.), Classification (C.), Detection (D.), and Segmentation (S.)

Paper	Objective	Architecture	Results
Srivastava et al. [9]	Detection of dental caries	Fully Connected NN	F1 _S : 70
Kumar et al. [10]	Segmentation of Dental Caries	U-net with	F1 _S : 61
Cantu et al. [11]	Segmentation of Dental Caries	U-Net	F1 _S : 79
Lian et al. [12]	Dental Caries Segmentation and Classification	nnU-Net, DenseNet121	IoU _S : 78.5, F1 _C : 0.790
Lee et al. [13]	Classification of Dental Caries	GoogLeNet Inception v3	F1 _C : 84.6
Silva et al. [14]	Tooth Segmentation and Numbering	PANet	F1 _N : 89.2, F1 _S : 88
Chen et al. [5]	Tooth Detection and Numbering	Faster R-CNN	mIoU _D : 91, F1 _N : 91.5
Muresan et al. [15]	Tooth segmentation and Classification	YOLO, ERFNet	F1 _S : 93
Chung et al. [16]	Tooth Detection and Numbering	Custom CNN	mIoU _D : 84

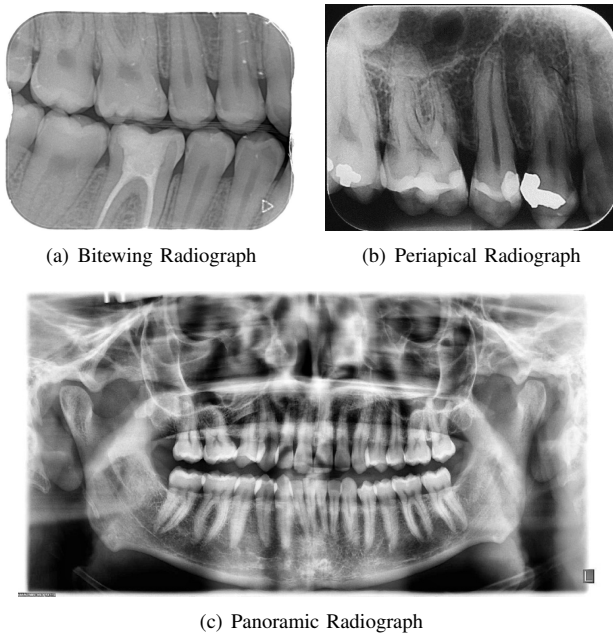


Fig. 1: Types of 2D radiography used for caries detection.

but also other anatomical structures like the maxillary and mandibular jaws and the sinuses [21].

The primary imperfections observed in most prior studies include their approaches' inability to effectively segment caries across various types of radiographs like panoramic and bitewing. Furthermore, most of them don't have independent datasets for testing purposes, and the size of the dataset is limited. The accuracy of segmentation and detection in these earlier models falls short of the standards required for medical applications. Finally, previous studies have tended to overlook the issue of class imbalance in radiography images, which is a natural feature of human teeth during the training process.

This research introduces a DL approach aimed at enhancing the precision of tooth feature diagnosis by enabling dentists to accurately segment caries in periapical, bitewing, and panoramic radiography images. To conduct this study, various datasets were utilized, encompassing complete panoramic X-ray scans and bitewing X-ray images. Moreover, a dataset of annotated bitewing images was created. The main con-

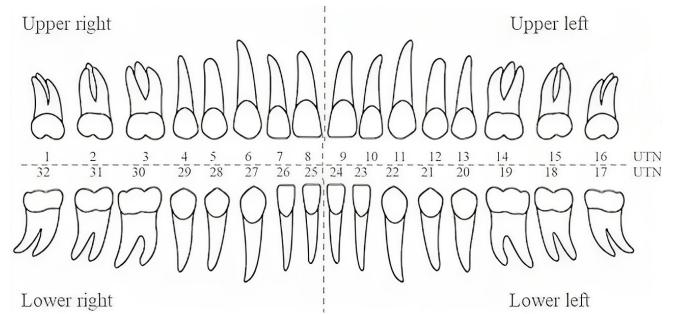


Fig. 2: UTN system is identified as an international standard [22].

tributions of our study are outlined below: To begin, the quality and quantity of the dataset have been improved through diverse transformations and filters. The second step involves employing an NN model for teeth detection. Subsequently, a new post-processing method is introduced, which classifies each tooth based on the universal teeth numbering pattern shown in Figure 2. The tooth detection model is fine-tuned using cropped panoramic X-ray images in the third step. It is a novel method of preparing the model for tooth extraction during further segmentation, comprising panoramic, bitewing, and periapical images. Finally, a segmentation model is trained on individual tooth images to predict caries and restoration segments.

The rest of the paper is organized into three sections. Section II describes our proposed method pipeline. Section III examines the experimental data and evaluates it. Finally, Section IV presents our concluding remarks.

II. MODEL

This section provides an overview of the YOLOv8 [23] model for detection and segmentation applications. YOLOv8 represents the latest advancement in the field of real-time object detection and segmentation, building upon the established reputation of its predecessors. This model exhibits unparalleled speed and accuracy in its operational capabilities. Moreover, YOLOv8 exhibits a notable advantage in terms of performance and network size compared to its previous versions. YOLOv8 is renowned for its remarkable customizability, enabling users

to effortlessly modify the model's architecture and settings to cater to their individual needs. In addition, YOLOv8 offers pre-trained models, which facilitate usage and enable the application of transfer learning and various datasets.

A. Model Architecture

The architecture of YOLOv8 has three main components: the *backbone*, *neck*, and *head*. The backbone serves as the fundamental framework of the network, and YOLOv8 presents a range of backbones, such as *EfficientNet*, *ResNet*, and *CSP-Darknet*, enabling users to customize the model according to their requirements. The neck serves as a connecting component between the backbone and the head. YOLOv8 integrates Spatial Pyramid Pooling—Fast (SPPF) and New CSP-PAN structures, which closely resemble the methodology employed in YOLOv5. Finally, YOLOv8 utilizes the head component to perform the output generation. In general, YOLOv8 has two heads. One is for segmentation tasks, and the other is for detection tasks [23].

B. Evaluation

This section will discuss the metrics employed to evaluate the model's performance. Precision is a frequently utilized metric in object detection and segmentation, indicating the model's ability to forecast positive predictions accurately. Precision is formally defined in Equation 1. In this equation, **TP** represents the true positive, and **FP** represents the false positive.

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

Recall is another important metric in object detection and segmentation. Recall refers to a model's ability to locate all relevant specific classes within a dataset. Recall is defined in Equation 2, where FN stands for false negative.

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

Another measure used to create a relationship between recall and precision is the F1 score, calculated as the harmonic mean of the precision and recall. The F1 score is mathematically specified in Equation 3.

$$F1 = \frac{2}{\frac{1}{Recall} + \frac{1}{Precision}} \quad (3)$$

One of the most important metrics in object detection and segmentation tasks is Intersection over Union (IoU). IoU measures the overlap between bounding boxes and polygons in the Grand Truth (GT) and predicted output. IoU is defined on Equation 4.

$$IoU = \frac{TP}{TP + FN + FP} \quad (4)$$

The final metric used in this study is mean Average Precision (mAP) which calculates the average precision score for each class or object category in a multi-class problem and then takes the mean of these scores.

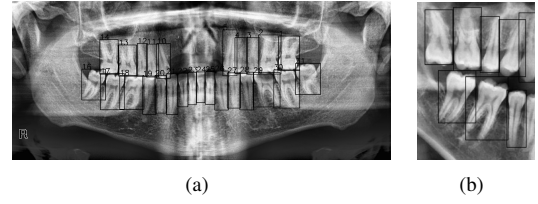


Fig. 3: Detection Results: a) is the detection and numbering output on a panoramic radiography b) is the detection and numbering output on a bitewing radiography.

III. METHODOLOGY AND MATERIALS

The main objectives of this research are to propose an NN-based algorithm to predict the position and number of individual teeth in the UTN system and segment caries and restoration proportions in full panoramic X-ray images. Furthermore, our method can detect and extract individual teeth in bitewing and periapical images.

The primary aim of cropping each tooth in the image for the segmentation process is to enhance the efficiency of data augmentation and improve the accuracy of predictions related to cavities and restorations. Another rationale for initially cropping each tooth is that this methodology enables our method to process any radiographic images mentioned in Figure 1.

The workflow of our method is organized into four distinct sections. In the initial section, a dataset of bitewing radiographs is annotated. Following this, the dataset is augmented and pre-processed to be prepared for training. The subsequent section focuses on teeth detection and numbering within panoramic images, followed by fine-tuning to detect teeth in bitewing radiography images.

The next step involves cropping every tooth of the original images and segmentation of cavities and restorations on each tooth. Finally, the integration of these modules is included. The training procedure utilizes the NVIDIA V100 Tensor Core and 16GB of RAM. During the training phase, the Adam optimizer with 0.9 momentum and the learning rate decay 0.0005 was employed. A complete listing of all training hyper-parameters is provided in Table II. The following subsections elaborate on comprehensive details concerning each of these sections.

TABLE II: Hyperparameters.

Model	Epochs	Batch Size	Image Size	Learning Rate
Teeth Detection and Numbering Training	100	32	640	0.00278
Teeth Detection and Numbering Fine-Tuning	20	32	640	0.00125
Cavity and Restoration Segmentation	100	64	480	0.01667

A. Datasets and Preparation

Our study utilizes two datasets to train and test models, detailed further below. The detection model is trained using the dataset provided by Tufts University of Dental Medicine [24]. The dataset contains 968 normal panoramic X-ray images

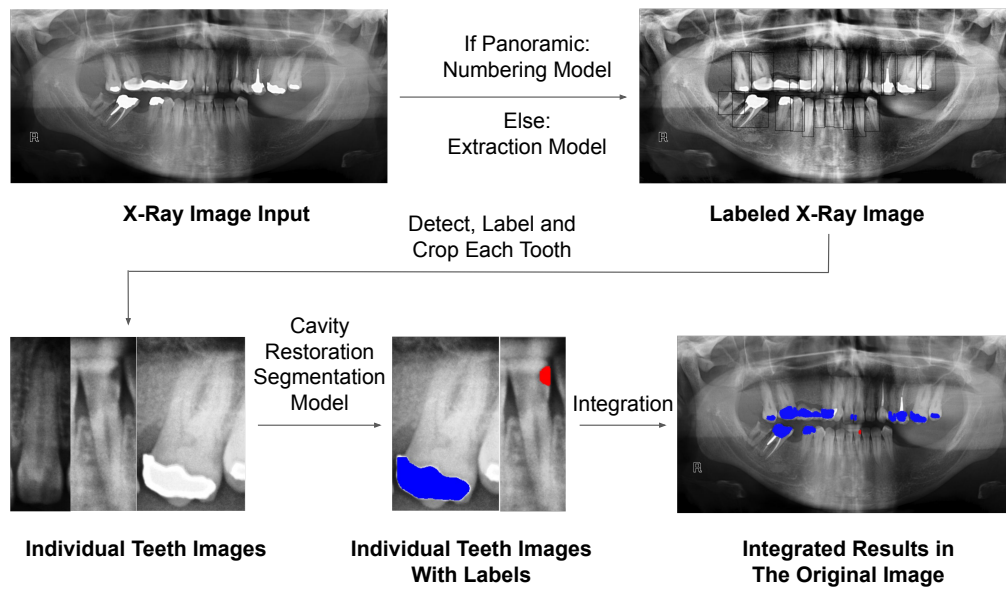


Fig. 4: Method flowchart

with bounding box coordination and number labels in the UTN system. This dataset is split into training and testing with a ratio of 80:20. Another dataset that was employed in this study is Tamami [25]. The Tamami dataset consists of 1085 bitewing radiographs with cavities and restorations segmentation labels. Finally, images of both datasets are enhanced using the CLAHE histogram equalizer filter, previously employed in [16], [26].

B. Teeth Detection and Numbering

This section explains the model's functionality and discusses the reason for devoting a model to identifying specific teeth. Moreover, our study developed a novel post-processing technique to propose precise teeth numbering and positioning. The YOLOv8 medium model is used in this study to detect each tooth in panoramic X-ray images. Augmentation was also applied to the dataset for training purposes. This step predicts a bounding box for each tooth, along with the assignment of the corresponding tooth number in the UTN system, as illustrated in Figure 2. The hyper-parameters for the training procedure are mentioned in the first row of Table II.

Regarding the logic behind post-processing, it should be mentioned that the symmetry of the human mouth creates a 50% probability that the model may confuse the left and right sides of the mouth during tooth detection since YOLOv8 is not a region-based model. To address this issue, a post-processing technique has been developed to mirror the predicted bounding boxes in case of side misclassification. Furthermore, in cases where certain teeth near the center of the image have the same labels, our post-processing technique facilitates sorting of the bounding boxes based on position.

After the model had trained on panoramic radiographs, a fine-tuning process was performed to detect the teeth in

bitewing images. Due to the lack of a publicly available dataset of bitewing radiographs with teeth annotations, panoramic radiographs and their corresponding annotations in the Tufts dataset were cropped to produce images resembling bitewing radiographs. The Tufts dataset's cropped images contain more incisors and canine teeth than molars and premolars. Therefore, a weighted augmentation approach has been applied to address this imbalance. The details of hyper-parameters for fine-tuning the model are mentioned in the Table II.

Finally, every predicted tooth from the detection model was cropped with a small overlap to be fed into the segmentation model later. To our knowledge, there isn't a suitable library for cropping images with their corresponding polygons. Thus, an algorithm has been developed to crop images along with their polygons in this study using the Albumentations Library.

C. Caries and Restoration Segmentation

In this section, two different models are trained for the caries and restoration segmentation task. The models' details and configurations are explained in subsection IV-B. A total number of 3971 individual teeth cropped from the Tamami dataset were utilized for training purposes. Additionally, the models were trained on the original dataset to compare with the cropping approach. To enhance the diversity of the dataset and address the imbalance in the distribution of classes, augmentation techniques involving stochastic horizontal and vertical flips were applied. The details of the training hyper-parameters are mentioned in the third row of Table II.

D. Integration: Proposed Unified Method

In the final step, previously explained modules were integrated into a unified method, resulting in a two-stage pipeline. This pipeline inputs a panoramic or bitewing X-ray image

and proceeds to predict the number and position of each tooth in the first stage. Then, the outcome is cropped and saved as new individual teeth X-ray images with correspondent labels. The process follows these cropped images to the second stage, where caries and restoration segments within individual tooth images are identified. In the last phase, an algorithmic procedure was developed to scale up the prediction of each tooth to match the original image and determine the absolute coordinates relative to the original image size. The predicted coordinates are between 0 and 1. Consequently, the method uses positions and the sizes (height and width) of bounding boxes to calculate the coordinates of polygons' predicted points on the original images. All the prediction processes use the optimum confidence of labels based on the maximum F1 score. As mentioned earlier, the F1 score indicated the balancing point of the precision-recall trade-off. Thus, this approach allows the pipeline to achieve the best performance of models.

IV. RESULTS

In summary, this paper presents a comprehensive analysis of the results encompassing the detection model, segmentation model, and the proposed unified method.

A. Teeth Detection and Numbering Results

Table III summarizes the performance of the NN models for teeth detection and numbering tasks. The YOLOv8 model achieved a precision of 71.8%, a recall of 59.0%, and a mAP50 of 72.0% for the bounding boxes on Tufts dataset. These results indicate a considerable amount of false positives and false negatives in the predictions of this model. Furthermore, the confusion matrix depicted in Figure 5 indicates that the model has confused and misclassified the teeth in the left and right half of the radiography image. For instance, teeth number 0 and 15 according to the UTN system are similar in shape and size but different in the position of the jaw. Therefore, the model could not distinguish the correct label. Consequently, the primary factor resulting in the low precision and recall is the inability of YOLOv8 to establish a meaningful correlation between the object's position in the image and its corresponding label. This is the result of the differences between YOLO and region-based networks.

A novel post-processing method has been proposed and applied to the algorithm to address this issue. This method's clarification is provided in subsection III-B. By employing this post-processing technique, precision, recall, and mAP50 improved by 21.5%, 17.6%, and 21.3%, respectively. Consequently, the proposed approach significantly boosted the detection model's performance.

Furthermore, the fine-tuning results on the Tufts cropped panoramic image dataset are presented in Table III. The post-processing technique was not employed because the cropped images and bitewing radiographs have no symmetry issues. The corresponding precision achieved by the model on this dataset is 96.9%, with a recall of 97.1%, and an mAP50 of 98.5%. In the following, the fine-tuned model on this dataset

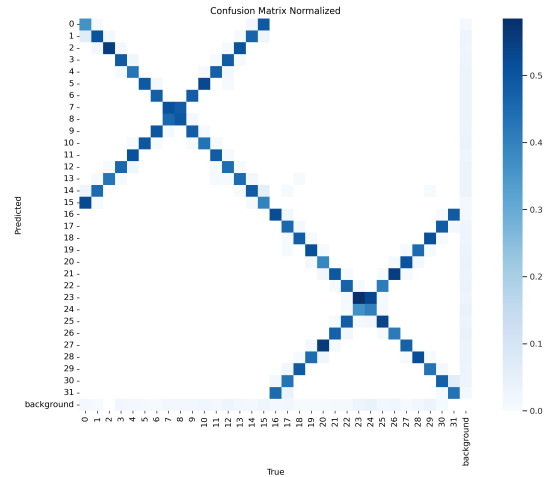


Fig. 5: Detection and Numbering model's Confusion matrix

is also evaluated on Tamami dataset. The precision of this evaluation was quantified as 95.4%, the recall as 81.6%, and the mAP50 as 88.0%. These results indicate that the fine-tuned detection model works well with bitewing images.

B. Teeth Cary and Restoration Segmentation Results

In this section, the performance of the segmentation models is evaluated. As mentioned, this task involves segmenting the cavity and restoring proportions in individual teeth (Tamami-cropped dataset). The results of two segmentation models, namely YOLOv8 and Detectron2 [27], are presented in this section. These models were evaluated using Tamami and Tamami-cropped datasets.

As highlighted in Table IV, YOLOv8 model results are better than Detectron2 in all metrics, datasets, and classes. For example, the YOLOv8 model has performed 9.7 and 13.9 percent better than the Detectron2 model in the average mAP50 on Tamami and Tamami-cropped datasets, respectively.

The Tamami dataset is imbalanced since only 25% of the 3225 data are in the Cavity class. Consequently, the model's performance in this particular class has been insufficient. The augmentation of the Tamami dataset is not feasible and leads to an increase in the number of instances for both classes. In the Tamami-cropped dataset, data augmentation can be performed due to the inclusion of individual images representing each tooth. By implementing this augmentation, the model's performance has been enhanced up to 15.7% for the Cavity class.

C. Proposed Unified Method Results

To evaluate our unified method, bitewing images from the Tamami datasets are utilized as input. The images are first fed into the detection module, with the precision of 95.4%, the recall of 81.6%, and the mAP@50 of 88.0% for finding bounding boxes around each tooth. Then, the teeth are cropped individually (with a safe margin) based on the proposed bounding boxes. These cropped images are entered into the segmentation module in the next step. The model achieved the

TABLE III: YOLOv8m Detection Models Results (with Post-process)

Application	Post-process	Dataset	Precision	Recall	mAP@50
Teeth Detection and Numbering	×	Tufts (Panoramic)	71.8%	59.0%	72.0%
Teeth Detection and Numbering	✓	Tuft (Panoramic)	93.3 %	76.6 %	93.6 %
Teeth Detection	×	Tufts (Cropped Panoramic)	96.9%	97.1%	98.5%
Teeth Detection	×	Tamami (Bitewing)	95.4%	81.6%	88.0%

TABLE IV: Caries and Restoration Segmentation Results

Model	Dataset	Precision		Recall		mAP@50	
		Cavity	Restoration	Cavity	Restoration	Cavity	Restoration
YOLOv8n-seg	Tamami	40.3%	76.9%	25.9%	85.5%	21.1%	87.6%
YOLOv8n-seg	Tamami-cropped	56.0%	87.8%	34.4%	79.3%	35.4%	89.5%
Detectron2	Tamami	2.39	55.19	1.8%	23.4%	2.41%	58.2%
Detectron2	Tamami-cropped	6.0%	52.2%	5.2%	26.7%	8.4%	55.3%

precision of 71.9%, the recall of 56.9%, and the mAP@50 of 62.4% as the metrics of the segmentation task. Finally, the segmented regions are scaled and mapped onto the original images. The overall results of our unified method are provided in the Table V by comparing the prediction output on the original images with the ground truth.

Our proposed method effectively combines detection and segmentation tasks, leveraging the benefits of analyzing teeth individually. Table V shows the performance of our proposed method and conventional segmentation. Precision witnessed an improvement of 0.7%, while recall demonstrated a substantial increase of 4.3%, and mAP@50 exhibited significant improvement by 6.8% compared to the state-of-the-art YOLOv8 model. This comparison demonstrates that, in addition to assisting in augmenting data for a specific class, our unified method has enhanced segmentation results.

TABLE V: Proposed Unified Method Results Comparison

Model	Precision	Recall	mAP@50
YOLOv8	57.7%	55.0%	54.3%
Our Method	58.4%	59.3%	61.8%

V. DISCUSSION AND CONCLUSION

Implementing a Computer Aided Diagnosis (CAD) System improves accuracy and dependability while minimizing complications caused by examiner differences. Our research presents a method for automatically detecting the number and position of teeth, cropping individual teeth from panoramic, bitewing, and periapical X-ray images, and segmenting dental caries and restoration on each tooth. In clinical routines, three important factors stand out: segmentation accuracy, cost, and time [8]. Manual segmentation is considered the gold standard due to dentists' knowledge and experience, while AI-based assistants can only propose suggestions regarding characteristics in radiographic images, they can also be game-changers in time and cost consumption [11]. They are available anytime and can provide services quickly from anywhere, making them cost-efficient and time-efficient. Another advantage is that their

performance is not affected by environmental parameters. Our method includes two DCNN modules corresponding to each mentioned task. The pipeline is completed by integration of all modules. Finally, an enhancement in the performance of cavity and restoration segmentation was achieved by our method. The results suggest that the two-stage approach will enable the algorithm to detect more cavities and restorations within the radiographs while increasing the accuracy of predictions.

In conclusion, this study employed a DCNN-based assistant to enhance dentists' diagnostic performance across dental caries and restorations by processing radiographs. This approach also reduces the time and costs associated with the healthcare process.

As the model was not region-based and post-process was needed to address the mislabeling problem, one of the future directions for developing this research is to modify YOLOv8 architecture to become region-based, with specific emphasis on the segmentation of cavities and restorations. The development of this specific DCNN is also needed to improve results with less computational cost. Furthermore, a noteworthy challenge is posed by the limited availability of datasets dedicated to cavities and restorations. Addressing this issue is important for the generalizability of DCNN models, demanding the acquisition of diverse data from multiple devices. Another avenue for future research pertains to the annotation of other oral characteristics.

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